Aegis Futures Lab — User Manual

For traders, not technicians. Everything you need to use the app, in plain language.

Live app: https://aegis-futures-lab-khaki.vercel.app This manual is also available inside the app on the **Guide** page.

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1. What this app is

Aegis watches the two micro futures markets — **MES** (Micro E-mini S&P 500) and **MNQ** (Micro E-mini Nasdaq-100) — and posts **practice trade ideas** two to three times a day using a demand-and-supply zone strategy. Every idea is tracked to its result (target hit, stop hit, or closed flat), so you can judge the strategy on evidence instead of memory.

> **Nothing here touches real money.** There is no broker connection, prices are delayed 10–15 minutes, and trade ideas appear 5–15 minutes after the setup happens. Use the app to practice, learn, and keep score — never as a live trade instruction.

2. Your daily routine

1. **Open Home with your morning coffee.** It is the screen the app starts on and it answers the whole morning in one look: how many ideas today (the dots fill toward the 2–3 per day target), today's profit or loss, when the bot checks next, and — at the top — the one idea that is live right now, with its entry, stop and target. If nothing is running, it says so plainly.
2. **Scroll on for the last three weeks.** One bar per trading day, green above the line and red below, with the net, the win rate and the number of ideas beside it. Below that: the two markets, the zones price is closest to, and whether the bot is healthy.
3. **Open Signals when you want the detail.** Every idea ever posted, grouped by day. Each row is a complete trade plan: where to get in (Entry), where the idea is wrong (Stop), where to take profit (Target), and how it ended (the Status badge).
4. **Glance at the Zone watchlist.** These are the buy and sell areas the strategy cares about, sorted by how close price is. An amber **AT ZONE** badge means price is sitting in one right now — the interesting moments happen there.
5. **After you trade, write it down.** On the Journal page, add your own trades by hand or import the CSV file your broker (Topstep / Tradovate) exports. The journal saves to the cloud automatically.
6. **On the weekend, keep score.** The Performance panel shows the win rate and running profit of each tier. Give the engine a few weeks of evidence before drawing conclusions — a handful of trades proves nothing, in either direction.

3. Tier A and Tier B — the two kinds of ideas

- **TIER A** — the classic zone setup: price returning to a fresh Daily or 4-hour demand/supply zone with everything lined up. These are **rare** (sometimes none for days) but they are the highest-conviction trades the strategy knows.

- **TIER B** — the daily bread-and-butter: a mean-reversion setup that fades short-term exhaustion, capped at two trades per market per day and shut off after two losses. These keep the feed active every day.

The point of the labels: over time, watch **which tier actually makes money** in the Performance panel, and weight your attention accordingly.

When the bot benches a strategy (circuit breakers)

The bot watches how each stream is doing. When a stream's recent results slump — profit factor below 0.8 over its last 20 finished trades — it **benches** that stream: it stops showing its ideas and stops counting them in the headline numbers, but keeps simulating them silently. When the silent practice recovers (profit factor 1.1 or better over the next 15), the bot returns the stream to the game on its own, waiting at least three trading days between changes so it never flip-flops. Paused streams appear in their own **Paused streams** box on Signals and a note on Home; the weekly digest also keeps their practice out of the headline numbers and reports it on its own line. Every bench and return is recorded and sent to Telegram. It is the safest kind of automation — learning when *not* to trade — and it is paper only.

What each idea tells you at a glance

Every idea on Home and on Signals carries four lines under it, so you do not have to hold the context in your head:

- **Setup** — what triggered it, in the strategy's own terms.
- **Invalidated** — the price that proves the idea wrong, and how far away it is. This is the stop: if price gets there, the reason for the trade is gone.
- **Odds** — the model's win probability for this idea, plus the zone score where there is one. If the model has not scored it, it says so rather than showing a number.
- **History** — how *this kind of setup* has actually done: the same tier in the same kind of market, and the same tier at the same level of market fear. **Every one of these comes with the number of trades behind it**, and when there are fewer than 10 it says "still collecting" and shows no rate at all. A 100% win rate on 3 trades is not information, and the app will not present it as if it were.

When nothing happens: "Why no signal today?"

A quiet day raises exactly one question — is the bot broken, or just being patient? The **Why no signal today?** box on Home answers it in a sentence, using the bot's own count of what it looked at: how many five-minute candles it checked, how many zones price actually reached, how many setups qualified, and what stopped the rest.

A big number of candles checked with nothing qualified is the **normal, healthy** state — this strategy is built to wait for price to come to it. The box also tells you when something else is going on: a strategy benched by the breaker, or a stalled price feed. If it cannot tell you why, it says so rather than guessing.

The model that learns to skip weak signals

A small model studies every signal the bot has seen and learns which setups are **least** likely to win. It can only ever do one thing: quietly skip the weakest 1-in-10 signals — it can never invent or enlarge a trade. It must **earn the right** to act: until it has at least 150 clean examples *and* beats a simple baseline on unseen data *two nights running*, it only shadow-votes (marking what it *would* have skipped, with the Saturday digest reporting how those would have done). The count it has and the count it needs are printed together everywhere it appears. If it graduates and later slips, it demotes itself. Its status, accuracy trend and calibration are on the **What the bot knows** page. Paper only.

The bot proposes its own upgrades

Once a week the bot searches for better strategy settings and tests them honestly — tuning on older data, checking on a held-out month it never saw, and stress-testing the worst-case drawdown. If the same improvement wins two weeks running (or a shadow strategy passes its promotion checklist two weeks running), it opens a **pull request** on GitHub with the evidence attached. A pull request is only a proposal: **nothing changes until you merge it**, the bot can never edit live settings itself, and it will not re-propose the same idea for a month. Most weeks it finds nothing and stays quiet. Merging is the one job left to you.

4. How to read one signal

Field	Meaning
Entry / Stop / Target	The full plan. Risk = entry to stop; reward = entry to target.
R:R	Reward-to-risk. 1.5 means the target pays 1.5× what the stop costs — you only need to win about 4 in 10 to come out ahead.
Status	**TARGET** = winner · **STOP** = loser · **OPEN** = still running · **CLOSED UP** = finished in profit without reaching a target (some strategies exit on their own signal rather than a fixed target; those wins used to be mislabelled as flat closes) · **FLAT CLOSE** = closed at 15:25 ET / 00:55 IST with nothing gained (the strategy never holds overnight).
P&L	Simulated dollars for the position size the engine chose (risking about $160 per trade), commissions already subtracted.
Regime	What kind of market the idea was born into: trending or ranging, quiet or volatile (e.g. **TR·HV** = trending, high volatility). It never changes the ideas — it is a label, so the Performance panel can show which conditions the strategy actually earns in.
Stale data	The feed is delayed 10–15 minutes by design, but sometimes it stalls for far longer. If the newest bar is more than **30 minutes** old when the bot runs, any idea it works out describes a market that has already moved on. Those ideas are still recorded — hiding them would hide the outage — but they are marked **STALE DATA**, kept out of every score, never sent to Telegram, and never used to teach the model. They appear in their own **Excluded: stale data** box on Signals.
Marginal / doubtful fill	An honesty check on the entry. The simulation assumes a resting order fills when price touches the entry level — in a real market a touch is often not enough. No chip = price traded cleanly through the level. **MARGINAL FILL** (amber) = price barely reached it but came back later. **DOUBTFUL FILL** (red) = price only kissed the level once; a real order likely never filled, so treat that idea's profit as imaginary. Every performance number is also restated "excluding doubtful fills".

5. What each page does

Page	What it's for
Home	The screen the app opens on. Today at a glance: the live idea, today's score, the last three weeks, the two markets, the nearest zones, **why there was no signal today**, and whether the bot is healthy.
Signals	Every idea, grouped by day, with the full zone watchlist and engine detail.
Markets	Delayed charts, a live strategy readout, and the news calendar — each week's high-impact U.S. events from a free live feed, backed by the official BLS and Fed schedules when the feed is down.
Journal	Pick any past day: see what the engine did, minute by minute, next to your own journaled trades. This is where the learning happens.
Strategy Lab	The workshop (advanced, optional). Change strategy settings and run backtests.
Compare / Data	More of the workshop — compare variants, load your own CSV history. Both sit under **More** in the side menu on a computer. The Data page also shows the app's own price archive (its five-minute history saved to the cloud daily, growing past the feed's 60-day limit) and the **Shadow lab**: four extra strategies auditioning silently on live data. Shadow results are **not signals** and never alert; a stream earns promotion interest only after ≥60 finished trades, PF ≥ 1.2, and profits in two different market regimes.
What the bot knows	Under **More** in the side menu. Every night the bot re-reads everything it has recorded and re-derives its own statistics — whether the zone score predicts winners, which market conditions each tier does well in, what the filters turn away, whether the fills still look believable, and how the shadow strategies are doing. Pure observation: nothing here is a trade idea and none of it changes what the bot does. Anything with too few finished trades reads "collecting (n=X of 10)".
Guide	The in-app version of this manual.

You never need the workshop pages to follow the signals.

6. ET or IST — your choice

Every time in the app can be shown on the New York exchange clock (**ET**) or on your own clock in India (**IST**). Use the **ET / IST** switch — bottom of the side menu on a computer, top right on a phone. The choice is remembered on that device and it changes every screen at once: signal times, the chart's time axis, the journal, the news calendar.

Two things deliberately do **not** move:

- **Trading days.** The blotter groups ideas by New York trading day, always. An idea posted at 21:20 IST belongs to that New York session, not to the next Indian date.

- **Journal entry times.** You type your own trades in ET, because that is what the chart and the engine's own timestamps use. When the app is set to IST, the form shows what your typed ET time means in IST as you go.

India does not change its clocks but the United States does, so the gap is 9½ hours from March to November and 10½ hours through the winter. The app works this out for you — the session rules always print both, like "flat by 15:25 ET (00:55 IST)", and that second figure shifts by itself when New York changes its clocks.

7. Put it on your phone

Open the site on your phone, then choose **Add to Home Screen** in the browser menu. It installs like an app and opens straight onto the Home screen, with the five main pages along the bottom.

7a. "Live vs tuning window" — is it still working?

The strategy's settings were chosen on past data, which promised a certain profit factor and pace for each stream. The Home panel **Live vs tuning window** compares that promise with what the live ideas have actually delivered since go-live. While a stream has fewer than 20 finished ideas it only says **collecting data** — a handful of trades proves nothing. After that: green **tracking** = reality matches the promise; amber **lagging** = earning less than promised but still above water; red **underwater** = the stream is losing money over a meaningful sample.

A red stream means **stop trusting that stream** — the market may have changed since the settings were tuned. It never means "trade harder to catch up". The muted line under each stream repeats the numbers excluding doubtful fills, the stricter honest version.

7b. Telegram alerts (optional)

The bot can message you on Telegram the moment an idea triggers — entry, stop, target and reward-to-risk, with the time in both ET and IST — and again when it closes with the result. These are **paper ideas, not orders**: the same delayed data and the same simulation as the app, just delivered to your phone. It also messages if an engine run fails, so a quiet feed means a healthy bot, not a broken one. (Set up once by the operator with a free Telegram bot; nothing to configure in the app.)

8. Words you'll see

Word	Meaning
Zone	A price area where big buying (demand) or selling (supply) showed up before. The strategy trades the return to these areas.
Fresh / Tested	Fresh = price hasn't come back to the zone yet (strongest). Tested = touched once already.
Paper trading	Practice trades with imaginary money. All trades in this app are paper trades.
Flat by 15:25 ET	The strategy closes everything before the New York session ends (00:55 IST in summer, 01:55 in winter). No overnight risk, ever.
Engine (the "bot")	The automated checker that re-reads the market every 15 minutes during London and New York hours. If Home or Signals says the bot is idle or a run failed, the feed is paused — not the market.
Win rate	Share of closed trades that made money.
Delayed data	Prices arrive 10–15 minutes late. Fine for studying, useless for live execution.
ET / IST	The two clocks the app can show. ET is New York exchange time — the clock the strategy is written in.

9. If something looks wrong

- **"Market closed — <holiday name>"** — a CME holiday (Good Friday, Thanksgiving, Christmas…). The bot rests on purpose and the day shows calmly as closed, not as a problem. On half days (MLK day, the day after Thanksgiving, Christmas Eve…) trading stops early — the session bar says "early close" and simulated positions are flat before the earlier bell.
- **"Data delayed more than usual"** — an amber note on Home (Bot status) and on the Signals heartbeat. The bot is running, but the prices it last saw are older than the usual 10–15 minutes (a slow feed or a missed check during market hours). Ideas simply catch up on the next pass — treat the current ones as extra-delayed.
- **"Bot idle" on Home / "Engine idle / stale" on Signals** — the scheduled checker missed its slot (it runs on a free scheduler that is sometimes 5–15 minutes late). It catches up on the next pass; nothing is lost, because every pass recomputes the full picture.
- **"Nothing open right now" on Home** — normal. Most of the day there is no live idea; the card tells you when the bot checks next.
- **No signals today** — quiet days happen, especially for Tier A. The pace dots simply stay empty. That is information too.
- **"Signal feed unreachable"** — your device is offline or the database is briefly unavailable. The page retries every minute on its own.

Manual version: matches the app as of 2026-07-25. If the app has changed since, the Guide page in the app is the up-to-date reference (this file is regenerated from it — see CLAUDE.md in the repository).